1. Tabela Verdade

Considere a proposição composta:

$$(P \vee Q) \to (\neg Q \wedge R)$$

Construa a tabela verdade completa para a proposição. (Lembre-se: número de linhas = 2^n, onde n é o número de proposições simples.)

Valor: 2,0 pontos

P	Q	R	$P \vee Q$	$\sim Q$	$\sim Q \wedge R$	$P \vee Q \to (\sim Q \wedge R)$
V	V	V	V	F	V	V
V	V	F	V	F	F	F
V	F	V	V	V	F	F
V	F	F	V	V	F	F
F	V	V	V	F	V	V
F	V	F	V	F	F	F
F	F	V	F	V	F	V
F	F	F	F	V	F	V

2. Operações com Matrizes

Dadas as matrizes:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 0 & 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ -2 & 2 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \\ 1 & 2 & 1 \end{pmatrix}$$

Calcule:

a) $3A - B + 2C$

b) $A^T \cdot B$

c) C^2 (ou seja, $C \cdot C$)

Valor: 2,0 pontos cada item

$$2C = 2 \times \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 & -2 \\ 4 & 0 & 6 \\ 2 & 4 & 2 \end{bmatrix}$$

a) $3A = 3 \times \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 0 & 2 & -1 \end{bmatrix} = \underbrace{\begin{bmatrix} 6 & 3 & 0 \\ -3 & 9 & 6 \\ 0 & 6 & -3 \end{bmatrix}}_{3A} - \underbrace{\begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix}}_{B} = \begin{bmatrix} 5 & 3 & -2 \\ -6 & 10 & 5 \\ -2 & 4 & -3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 2 & -2 \\ 4 & 0 & 6 \\ 2 & 4 & 2 \end{bmatrix}}_{2C} = \underbrace{\begin{bmatrix} 5 & 5 & -4 \\ -2 & 10 & 11 \\ 0 & 8 & -1 \end{bmatrix}}_{3A-B+2C}$$

b) $\underbrace{\begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & -1 \end{bmatrix}}_{A^T} \times \underbrace{\begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix}}_{B}$

$= 2\times1+(-1)\times3+0\times(-2),\ 2\times0+(-1)\times(-1)+0\times2,\ 2\times2+(-1)\times1+0\times0$

$1\times1+3\times3+2\times(-2),\ 1\times0+3\times(-1)+2\times2,\ 1\times2+3\times1+2\times0$

$0\times1+2\times3+(-1)\times(-2),\ 0\times0+2\times(-1)+(-1)\times2,\ 0\times2+2\times1+(-1)\times0$

$= 2-3+0,\ 0+1+0,\ 4-1+0$

$1+9-4,\ 0-3+4,\ 2+3+0$

$0+6+2,\ 0-2-2,\ 0+2+0$

$$= \begin{bmatrix} -1 & 1 & 3 \\ 6 & 1 & 5 \\ 8 & -4 & 2 \end{bmatrix}$$

C^2

c) $\begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \\ 1 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \\ 1 & 2 & 1 \end{bmatrix} =$

$0\times0+1\times2+(-1)\times1,\ 0\times1+1\times0+(-1)\times2,\ 0\times(-1)+1\times3+(-1)\times1$

$2\times0+0\times2+3\times1,\ 2\times1+0\times0+3\times2,\ 2\times(-1)+0\times3+3\times1$

$1\times0+2\times2+1\times1,\ 1\times1+2\times0+1\times2,\ 1\times(-1)+2\times3+1\times1$

$= 0+2-1,\ 0+0-2,\ 0+3-1$

$0+0+3,\ 2+0+6,\ -2+0+3$

$0+4+1,\ 1+0+2,\ -1+6+1$

$$= \begin{bmatrix} 1 & -2 & 2 \\ 3 & 8 & 1 \\ 5 & 3 & 6 \end{bmatrix}$$

3. Determinantes e Inversas

Com relação às matrizes do exercício anterior, calcule:

a) Determinante de C

b) Matriz inversa de B e de A, caso existam

Valor: 2,0 pontos cada item

a) $\begin{matrix} 0 & 1 & -1 \\ 2 & 0 & 3 \\ 1 & 2 & 1 \end{matrix} \ \vdots \ \begin{matrix} 0 & 1 \\ 2 & 0 \\ 1 & 2 \end{matrix}$

$0\times0\times1=0$
$1\times3\times1=3$
$-1\times2\times2=-4$
$3-4=-1$

$1\times2\times1=2$
$0\times3\times2=0$
$-1\times0\times1=0$
2

$-1-2=-3$

b) $B^{-1}= \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix} \begin{vmatrix} 1 & 0 \\ 3 & -1 \\ -2 & 2 \end{vmatrix}$

$1\times(-1)\times0=0$
$0\times1\times(-2)=0$
$2\times3\times2=12$
12

$0\times3\times0=0$
$1\times1\times2=2$
$2\times(-1)\times(-2)=4$
6

$12 - 6 = 6$

~~1 0 2 1 0~~
3 -1 1 3 -1
-2 2 0 -2 2
1 0 2 1 0
3 -1 1 3 -1

$-1\times0-2\times1=-2$
$2\times2-0\times0=4$
$0\times1-(-1\times2)=2$

$1\times(-2)-6\times3=-2$
$0\times1-2\times(-2)=4$
$2\times3-1\times1=5$

$3\times2-(-2\times(-1))=4$
$-2\times0-1\times2=-2$
$1\times(-1)-3\times0=-1$

$$\frac{1}{6} \begin{bmatrix} 2 & 4 & 2 \\ -2 & 4 & 5 \\ 4 & -2 & -1 \end{bmatrix} = \begin{bmatrix} \frac{2}{6} & \frac{4}{6} & \frac{2}{6} \\ \frac{-2}{6} & \frac{4}{6} & \frac{5}{6} \\ \frac{4}{6} & \frac{-2}{6} & \frac{-1}{6} \end{bmatrix}$$

$A^{-1}=$

$\begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 0 & 2 & -1 \end{bmatrix} \begin{vmatrix} 2 & 1 \\ -1 & 3 \\ 0 & 2 \end{vmatrix}$

$2\times3\times1=-6$
$1\times2\times0=0$
$0\times(-1)\times2=0$
-6

$1\times(-1)\times(-1)=1$
$2\times2\times2=8$
$0\times3\times0=0$
$8+1=9$
9

$-6 - 9 = -15$

~~2 1 0 2 1~~
-1 3 2 -1 3
0 2 -1 0 2
2 1 0 2 1
-1 3 2 -1 3

$3\times(-1)-2\times2=-7$
$2\times0-1\times(-1)=1$
$1\times2-3\times0=2$

$2\times0-(-1)\times(-1)=-1$
$-1\times2-0\times0=-2$
$0\times(-1)-2\times2=-4$

$-1\times2-0\times3=-2$
$0\times1-2\times2=-4$
$2\times3-(-1)\times1=7$

$$-\frac{1}{15} \times \begin{bmatrix} -7 & 1 & 2 \\ -1 & -2 & -4 \\ -2 & -4 & 7 \end{bmatrix} = \begin{bmatrix} \frac{7}{15} & -\frac{1}{15} & \frac{-2}{15} \\ \frac{1}{15} & \frac{2}{15} & \frac{4}{15} \\ \frac{2}{15} & \frac{4}{15} & \frac{-7}{15} \end{bmatrix}$$

4. Método de Eliminação de Gauss

Utilize o Método de Eliminação de Gauss para resolver o sistema:

$$\begin{cases} x + 2y - z = 4 \\ 2x - y + 3z = 1 \\ 3x + y + 2z = 7 \end{cases}$$

Valor: 2,0 pontos

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & -1 & 3 & 1 \\ 3 & 1 & 2 & 7 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & -5 & 5 & -7 \\ 3 & 1 & 2 & 7 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & -5 & 5 & -7 \\ 0 & -5 & 5 & -5 \end{array}\right]$$

$\frac{-5}{-5} = \frac{5}{5} = 1$

$L_2 = L_2 - 2 \times L_1$
$2-2\times1=0$
$-1-2\times2=-5$
$3-2\times(-1)=5$
$1-2\times4=-7$

$L_3 = L_3 - 3 \times L_1$
$3-3\times1=0$
$1-3\times2=-5$
$2-3\times(-1)=5$
$7-3\times4=7-12=-5$

$L_3 = L_3 - 1 \times L_2$
$-5-1\times(-5)=-5+5=0$
$5-1\times5=5-5=0$
$-5-1\times(-7)=-5+7=2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & -5 & 5 & -7 \\ 0 & 0 & 0 & 2 \end{array}\right]$$ Impossível

5. Transformação Linear

Sabendo que $T : \mathbb{R}^2 \to \mathbb{R}^2$ é uma transformação linear e que

$$T(1,0) = (3,1), \quad T(0,1) = (-2,4)$$

Determine $T(x,y)$.

Valor: 2,0 pontos

$a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

$1a + b0 = x \iff a = x$
$0a + b1 = y \iff b = y$

$x\begin{bmatrix} 3 \\ 1 \end{bmatrix} + y\begin{bmatrix} -2 \\ 4 \end{bmatrix}$

1ª coordenada
$3x + (-2y)$
$3x - 2y$

2ª coordenada
$1x + 4y$

$T(1,0) = (3\times1-2\times0,\ 1\times1+4\times0) = (3,1)$ ✓
$T(0,1) = (3\times0-2\times1,\ 1\times0+4\times1) = (-2,4)$ ✓

6. Independência Linear

Verifique se os vetores

$$v_1 = (1,2,-1), \quad v_2 = (3,1,4), \quad v_3 = (0,5,2)$$

são linearmente independentes em $\mathbb{R}^3$.

Valor: 2,0 pontos

$\begin{bmatrix} 1 & 3 & 0 \\ 2 & 1 & 5 \\ -1 & 4 & 2 \end{bmatrix} \begin{matrix} 1 & 3 \\ 2 & 1 \\ -1 & 4 \end{matrix}$

$1\times1\times2=2$
$3\times5\times(-1)=-15$
$0\times2\times4=0$
$2-15=-13$

$3\times2\times2=12$
$1\times5\times4=20$
$0\times1\times(-1)=0$
$12+20=32$

$-13-32=-45$